Deep Shah

Ahmedabad, Gujarat, India · dpshah2307@gmail.com · +91 6352972571 · github.com/dpshah23 · linkedin.com/in/dpshah23 · leetcode.com/u/dpshah2307

SUMMARY

Computer Science undergraduate (B.Tech, graduating 2028) focused on backend development in Python and Django, with Java on Android. Built a Stripe payment service with idempotent requests, verified webhooks and a double-entry ledger, and added Razorpay payments during an internship. Published a paper on epidemic modelling.

TECHNICAL SKILLS

Languages	Python, Java, C++, SQL, JavaScript
Core CS	Data Structures, Algorithms, Object-Oriented Design, DBMS, Operating Systems
Backend & Web	Django, REST APIs, Stripe API, Razorpay, Webhooks, PostgreSQL, MySQL
Android	Java, Firebase, Android Keystore, BiometricPrompt
Tools	Git, GitHub Actions, Docker, Linux, CI/CD, Automated Testing

EXPERIENCE

Software Development Engineer Intern – Technostacks June 2024 – Aug 2024

Ahmedabad, India

- Built Django backend features for web applications; changes went through peer code review.
- Added Razorpay payments with webhook signature checks and transaction handling.
- Profiled slow PostgreSQL queries and added indexes, cutting API response time by 30%.
- Moved services to Docker and set up CI/CD, cutting environment setup from about 2 hours to under 5 minutes.

PROJECTS

LedgerPay – Stripe Payment Service Python, Django, PostgreSQL, Stripe API, GitHub Actions

- Creates Stripe PaymentIntents with a unique idempotency key, so a retry returns the original payment instead of charging twice; a failed Stripe call rolls back the database insert so the key can be retried.
- Webhook handler verifies Stripe signatures, skips duplicate event IDs, ignores out-of-order status changes, and writes ledger debit and credit entries in the same transaction as the status update.
- Tests cover duplicate requests, 20 concurrent requests on one key, replayed webhooks and out-of-order events, and run on PostgreSQL in GitHub Actions.
- Used AI tools (GitHub Copilot) for first drafts, then reviewed the code and confirmed each test fails when the logic it covers is removed.

CipherFlow – End-to-End Encrypted File Sharing Java, Android, Firebase, Android Keystore

- Each transfer uses its own AES-256-GCM key, wrapped with the receiver's RSA-2048 key (OAEP); the sender's SHA256withRSA signature is checked before decryption.
- Private keys stay in Android Keystore; decryption needs biometrics, previews stay in memory, and files can expire or delete after one open.

Invyra – Invoice Processing Pipeline Python, REST APIs, Task Queues

- Background services accept invoice documents over HTTP, parse them and extract fields.
- Task queues process invoices concurrently, with error handling for failed jobs; includes API tests and documentation.

RESEARCH

Published Author – CISTM 2025 (CRC Press)

- Multi-Wave Comparative Study of COVID-19 Transmission in India Using Extended SEIR and SEIRVD Models; fitted model parameters to real-world data.

EDUCATION

Nirma University, Ahmedabad Aug 2025 – May 2028

B.Tech, Computer Science Engineering CGPA: 8.08/10

Government Polytechnic, Ahmedabad Aug 2022 – May 2025

Diploma, Computer Engineering CGPA: 9.0/10

ACHIEVEMENTS & LEADERSHIP

- First Time Hacker Prize Winner – Spark the Summer Hackathon (2024)
- Executive Committee Member, Computer Society of India, Nirma University (Jan 2026 – Present)
- Executive Committee Member, Cyber Security Club, Nirma University (Oct 2025 – Present)